



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A)

A.Y. 2026 - 2027

TITLE OF THE PROJECT	Comparative Analysis of Time Series Forecasting Models Across Different Demand Characteristics	
1		
2		
3		
4		
Name of the Guide		
Name of the Guide		

Abstract

Demand forecasting is an important time-series analytics task used in areas such as retail, inventory planning, production, energy management, transportation, and resource allocation. It involves estimating future demand from historical observations so that organizations can make informed operational decisions. However, demand time series are not uniform in nature. Different series may exhibit long-term trends, recurring seasonal patterns, intermittent demand, sudden fluctuations, changing variability, and different levels of temporal dependence. These characteristics can influence the predictability of the series and the effectiveness of a forecasting model. Therefore, selecting a forecasting approach only on the basis of its overall performance may not provide sufficient understanding of its suitability for a particular type of demand behaviour.

Recent research has proposed a wide range of forecasting approaches, from established statistical methods to machine-learning, deep-learning, and newer foundation-model-based approaches. Benchmarking studies have also shown that forecasting performance can vary substantially across datasets, forecasting settings, and evaluation procedures. This project therefore proposes a comparative analysis of selected established time-series forecasting models across benchmark demand-oriented datasets representing different demand characteristics. The study will begin with exploratory analysis of the selected time series to identify characteristics such as trend, seasonality, intermittency, and variability. A common chronological data-splitting and forecasting procedure will then be used to train and evaluate the selected models, while suitable baseline methods will be included to provide meaningful reference points. Model performance will be assessed using appropriate forecasting error measures under a consistent evaluation framework.

The major focus of the project is not simply to identify one model with the lowest overall forecasting error, but to investigate how the relative performance and suitability of different forecasting approaches changes with the characteristics of the demand series. The analysis will compare the models across multiple benchmark datasets and examine whether specific demand behaviours are associated with better performance from particular forecasting approaches. The study will use publicly available benchmark data and standard data-analysis and forecasting tools, making the proposed work reproducible and feasible for systematic experimentation. The expected outcome is a structured empirical analysis that provides evidence-based insights into model suitability for different demand characteristics, together with a reproducible forecasting-analysis workflow that can be further extended in subsequent stages of the project.

Signature of the Guide

HOD